

Unit 6 Practice — Functions & Linear Relationships

Functions · Slope · Tables → Equations · Real-World · Comparing · Qualitative Graphs · Systems

A. Functions

Problem 1

Does the table shown represent a function? (yes/no)

x	y
1	5
2	7
3	9
4	11

yes

Problem 2

Does the table shown represent a function? (yes/no)

x	y
2	4
3	6
2	9
5	8

no

Problem 3

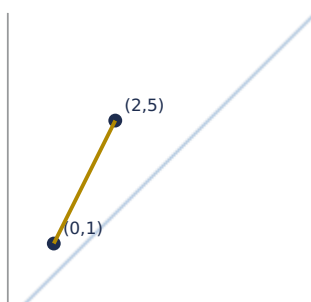
A relation includes the points (1,3), (2,5), (3,7), (1,9). Is it a function?

no

B. Slope

Problem 1

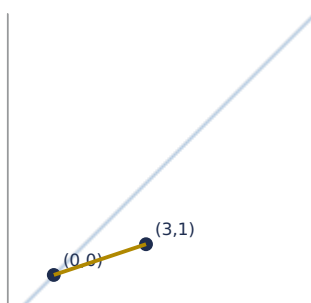
Find the slope of the line through the two points shown.



2

Problem 2

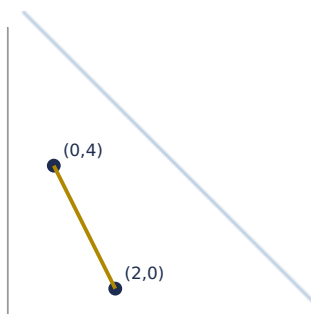
Find the slope of the line through the two points shown.



1/3

Problem 3

Find the slope of the line through the two points shown.



-2

Problem 1

Write the equation ($y = mx + b$) for the table shown.

x	y
0	1
1	3
2	5
3	7

$y = 2x + 1$

Problem 2

Write the equation ($y = mx + b$) for the table shown.

x	y
0	10
1	7
2	4
3	1

$y = -3x + 10$

D. Real-World Applications**Problem 1**

A taxi charges a \$4 flat fee plus \$2 per mile. Write the cost equation ($y = mx + b$) where x = miles.

$y = 2x + 4$

Problem 2

A gym charges \$30 to join plus \$25/month. How much for 6 months total?

180

E. Comparing Functions**Problem 1**

Function A: $y = 3x + 2$. Function B has a table showing slope 5. Which has the greater rate of change?

function b

Problem 2

Function A: $y = 4x - 1$. Function B passes through (0,3) and (2,7). Which has the greater y-intercept?

function b

F. Qualitative Graphs**Problem 1**

A graph shows temperature rising steadily, then staying flat for an hour, then rising again. During the flat part, is temperature increasing, decreasing, or constant?

constant

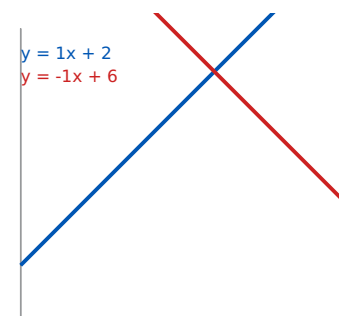
Problem 2

A car's distance-time graph shows a steep upward line, then a horizontal line. What is happening during the horizontal part?

stopped

G. Systems of Equations**Problem 1**

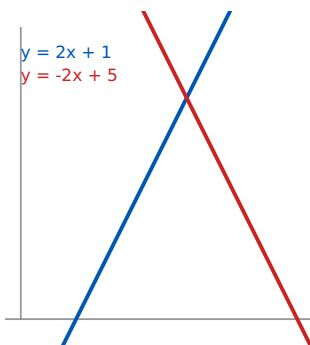
Find the solution (x,y) to the system of equations graphed.



(2,4)

Problem 2

Find the solution (x,y) to the system of equations graphed.



(1,3)

Problem 3

Solve the system: $y = x + 2$ and $y = 3x - 2$. Find x .

2